A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Captive breeding, habitat creation & management - Set 2

Mark scheme

AQA spec 3.1.2.3.3-3.1.2.3.4; The living environment; textbook pp. 32-49

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **In-situ:** species remain in natural conditions with the whole community and inter-species relationships – protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • **Limits of in-situ:** external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • **Ex-situ:** captive breeding when in-situ is impossible or insufficient – selection criteria – difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation) – techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning) – seed banks – successes (Hawaiian goose, Iberian lynx, red kite, large blue) • **Limits of ex-situ:** expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • **Judgement:** ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to – both needed, in-situ fundamental **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	9
	• **plagioclimax** → a community held back from its climax by human activity **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	
	Model answer - Level 3 (8 marks)	

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Question	Marking guidance	Marks
	In-situ conservation keeps a species in its natural habitat with the whole community it depends on - food species, pollinators, predators and the abiotic conditions it is adapted to. It is delivered through protected areas such as SSSIs, NNRs, SACs, SPAs, Ramsar sites and Marine Conservation Zones, land ownership by conservation bodies, agri-environment schemes and habitat creation and management. Its weaknesses are that external threats such as pollution and climate change cross reserve boundaries, enforcement and cost, conflicts with other land uses, and the fact that many populations are already small and fragmented.	
	Ex-situ captive breeding is used when in-situ protection is impossible or insufficient. It is hard: animals need enough space, the right food, social groupings and breeding triggers, and small founder populations risk inbreeding and hybridisation, which studbooks, artificial insemination and embryo transfer help to manage. It is expensive, works for few species, and captive animals may lose wild behaviour; release succeeds only where suitable habitat exists and with soft release and post-release monitoring.	
	The two are not alternatives. Ex-situ methods are a temporary safeguard that can rescue a species from extinction, as with the Arabian oryx, but they only succeed if there is protected habitat to return to. In-situ conservation is therefore fundamental and ex-situ a support for it; the relative importance depends on how close to extinction the species is.	
	Weaker answer - Level 2 (5 marks)	
	In-situ conservation means protecting species in their natural habitat, for example in nature reserves and national parks. This is good because the species stays in the wild with the other species it needs. Ex-situ conservation means captive breeding in zoos and then releasing the animals back into the wild. This can save species that are nearly extinct but it is expensive and animals may not survive when released. Both methods are important and are often used together.	
	What makes the difference	
	• Level 2 defines both approaches correctly but Level 3 gives specific mechanisms (types of protected area, breeding difficulties, release methods) and named examples.	
	• Level 3 weighs the limitations of each approach explicitly; Level 2 gives one limitation for ex-situ only.	
	• 'Both are important' is not a judgement about *relative* importance; Level 3 explains why in-situ is fundamental and when ex-situ matters most.	
01.2	1 mark for the point below. **1** Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

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